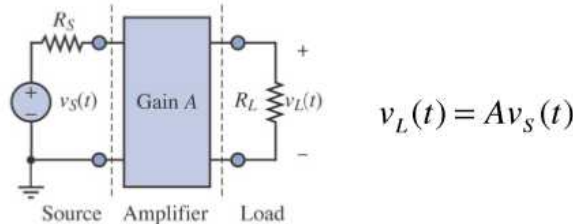


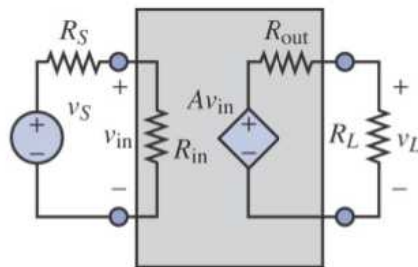
Chap.5 The Operational Amplifier

§5.0 Ideal Amplifiers

- Ideal Amplifiers: amplification of input signal to output signal with a certain amplification ratio.



- The internal structure of the amplifier can also be represented with Thevenin's equivalent circuit.

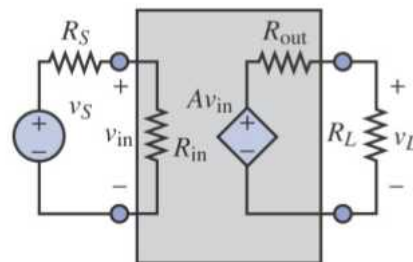


Ideal Amplifiers

$$v_{in} = \frac{R_{in}}{R_S + R_{in}} v_S$$

$$v_L = A v_{in} \frac{R_L}{R_L + R_{out}}$$

$$= \left(A \frac{R_L}{R_L + R_{out}} \frac{R_{in}}{R_S + R_{in}} \right) v_S$$



- ➡ • The amplification factor is dependent on both the source and load impedance
- If the input resistance R_{in} is very large & output resistance R_{out} is very small,

$$v_L \approx \left(A \frac{R_L}{R_L} \cdot \frac{R_{in}}{R_{in}} \right) v_S \approx A v_S$$

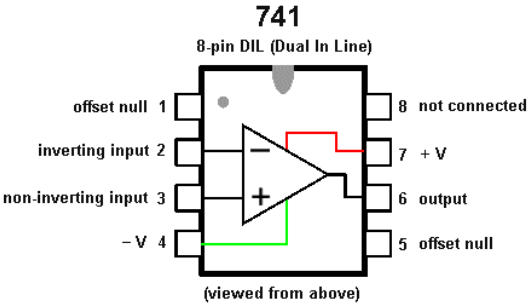
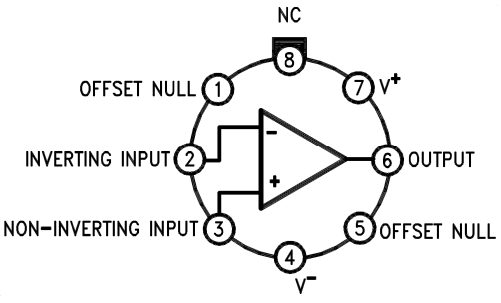
- ➡ Amplification factor is independent of source and load impedances.

§5.1 The Operational Amplifier (op amp)

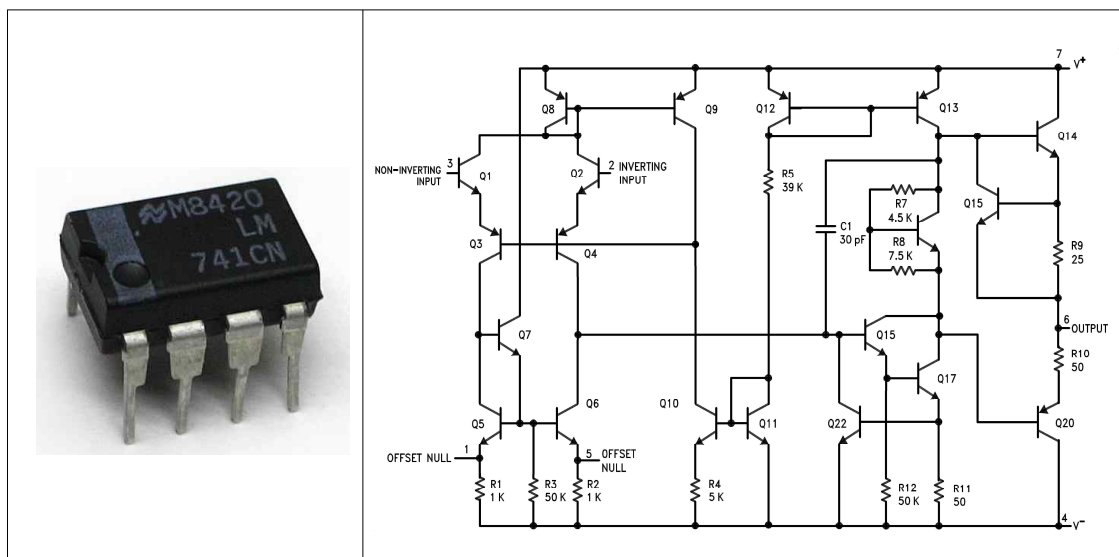
- Integrated Circuit that can perform a number of operations:
addition, amplification, differentiation, integration, filtering,
- ※ Analog Computer / Digital Computer
- ※ Linear IC

- LM741 Operational Amplifier

(<https://www.ti.com/lit/ds/symlink/lm741.pdf>)

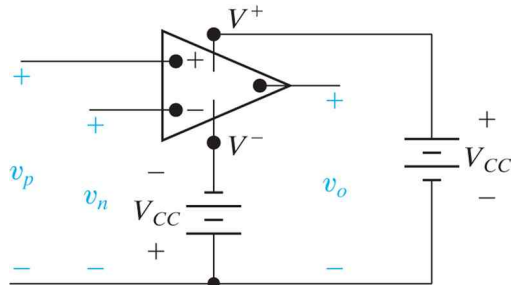
DIP package type	TO-99 package type
 741 8-pin DIL (Dual In Line) offset null 1 inverting input 2 non-inverting input 3 - V 4 8 not connected 7 + V 6 output 5 offset null (viewed from above)	 NC 8 OFFSET NULL 1 INVERTING INPUT 2 NON-INVERTING INPUT 3 4 - V 5 OFFSET NULL 6 OUTPUT 7 V+

- LM741 Operational Amplifier IC and the internal circuit



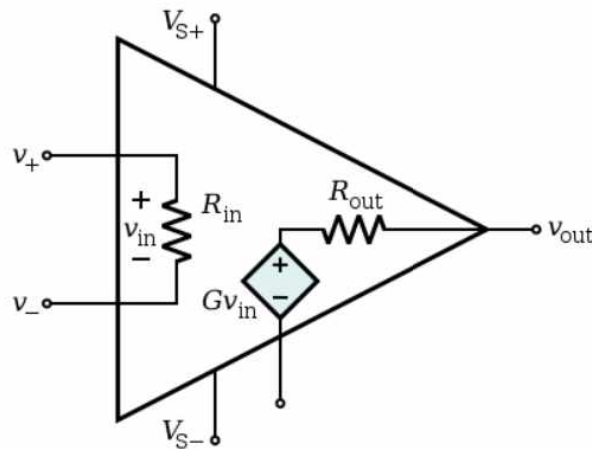
§5.2 Terminal Voltages and Currents

- Symbol of operational amplifier



v_p : noninverting input
 v_n : inverting input
 v_o : output
 V^+ : positive power supply
 V^- : negative power supply

- Ideal operational amplifier



1. Ideal op amp behaves as an ideal difference amplifier (amplifies the difference between two input voltages).

$$v_o = A(v_p - v_n)$$

2. near-infinite input resistance

$$R_{in} = \infty$$

3. near-zero output resistance

$$R_{out} = 0$$

4. Due to near-infinite input resistance, it is usually assumed that the input current is zero.

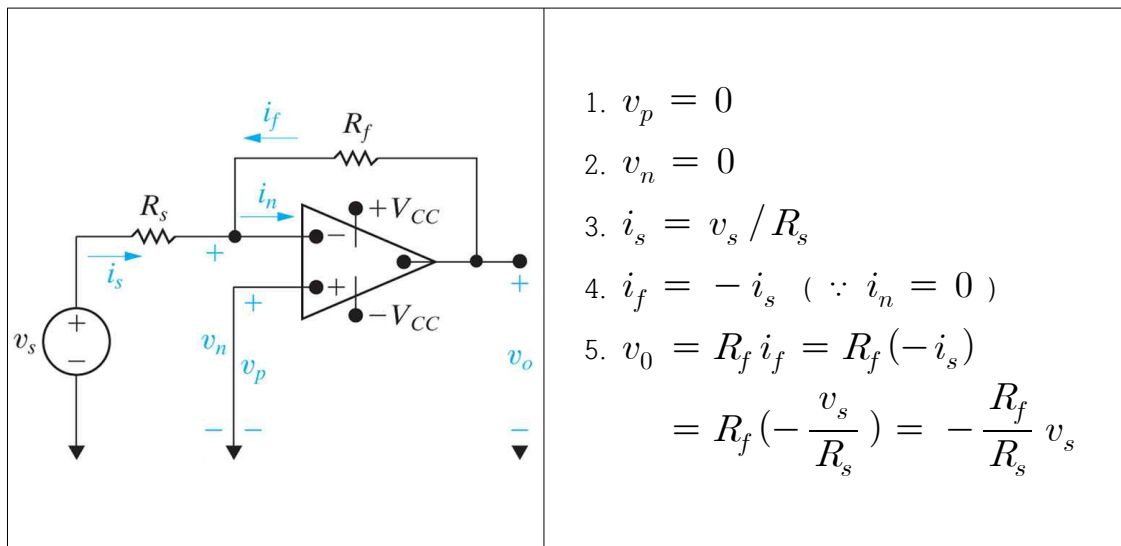
$$i_p = i_n = 0$$

5. Due to near-infinite open-loop gain ($A = \infty$)

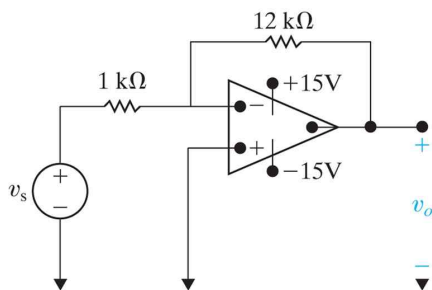
$$v_p = v_n$$

(※ actual op amp : $A = 10^5 \sim 10^7$)

§5.3 The Inverting-Amplifier Circuit



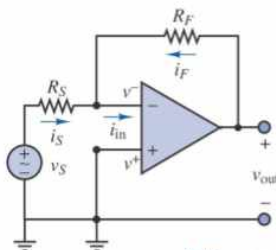
(Example) Inverting Amplifier



$$v_o = -12 v_s$$

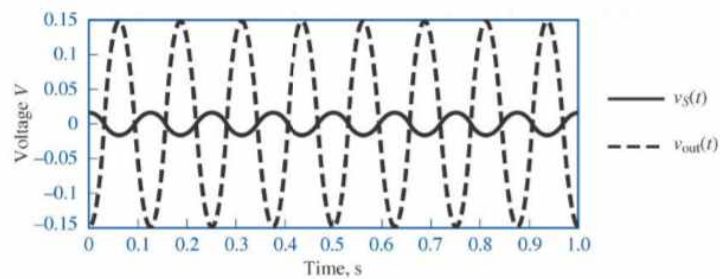
(Example) Inverting Amplifier

- If $R_s = 1\text{k}\Omega$, $R_f = 10\text{k}\Omega$, and $v_s = 0.015\cos(50t)$



$$v_{out} = A_v v_{in} = -\frac{R_f}{R_s} v_{in}$$

$$= -10 \times 0.015 \cos(50t) = -0.15 \cos(50t)$$



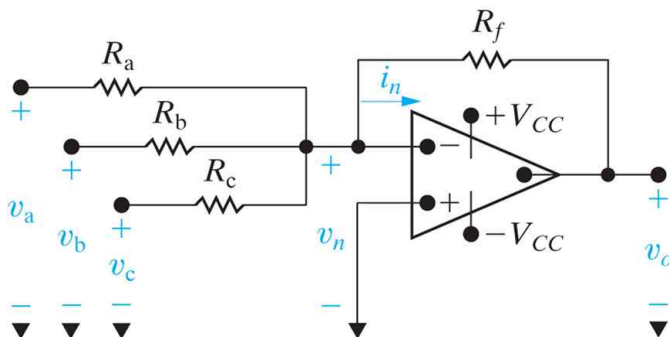
- closed loop and open loop

closed loop (※ feedback)	open loop
overall gain: smaller stability: higher	gain: very high stability: lower (∴ not used)

※ Negative feedback systems are widely used in control systems.

§5.4 The Summing-Amplifier Circuit

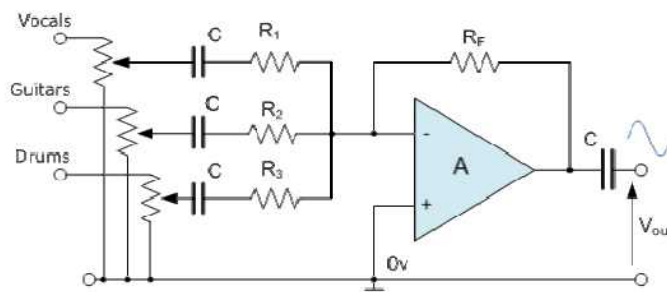
[※ Adder in analog systems]



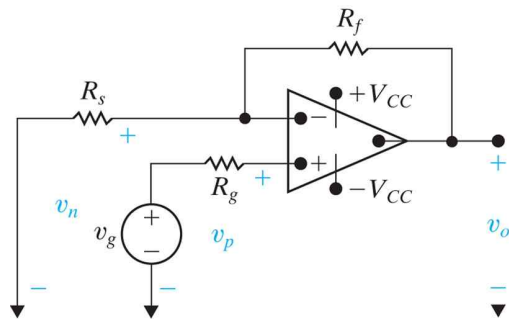
$$v_o = - \left(\frac{R_f}{R_a} v_a + \frac{R_f}{R_b} v_b + \frac{R_f}{R_c} v_c \right)$$

(※ 4-bit binary full adder)

(Example) Summing Amplifier : Audio Mixer

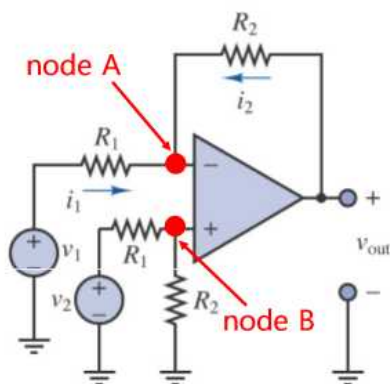


§5.5 The Noninverting-Amplifier Circuit



$$v_0 = \frac{R_s + R_f}{R_s} v_g = \left(1 + \frac{R_f}{R_s} \right) v_g$$

§5.6 The Difference Amplifier Circuit



- KCL at node A,

$$i_1 + i_2 = \frac{v_1 - v^-}{R_1} + \frac{v_{out} - v^-}{R_2} = \frac{v_1 - v^+}{R_1} + \frac{v_{out} - v^+}{R_2} = 0$$

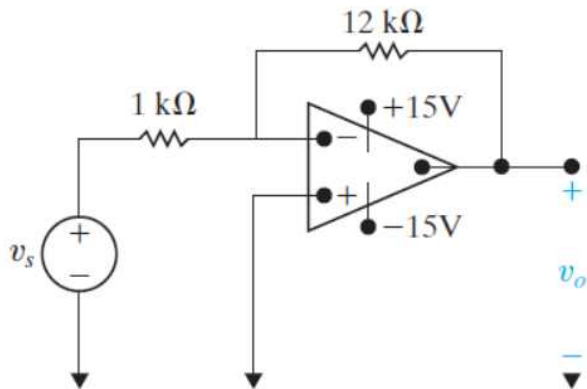
- At node B,

$$v^+ = \frac{R_2}{R_1 + R_2} v_2$$

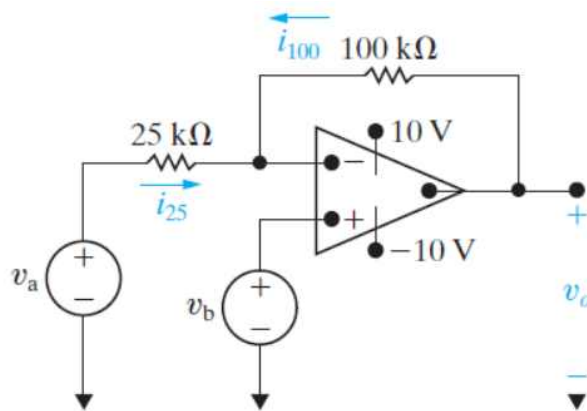
$$\begin{aligned} \Rightarrow \frac{v_{out}}{R_2} &= \left(\frac{1}{R_1} + \frac{1}{R_2} \right) v^+ - \frac{1}{R_1} v_1 \\ &= \left(\frac{1}{R_1} + \frac{1}{R_2} \right) \frac{R_2}{R_1 + R_2} v_2 - \frac{1}{R_1} v_1 = \frac{1}{R_1} (v_2 - v_1) \end{aligned}$$

$$\Rightarrow \boxed{v_{out} = \frac{R_2}{R_1} (v_2 - v_1)} \quad \text{Differential amplifier closed-loop gain}$$

(Example) v_o ?



(Example) v_o ?



(Example) v_o ?

